

From Robust Fusion to Statistically Valid Monitoring: An Anytime-Valid Conformal Layer for Detecting Real-World Weather-Onset Degradation in Multimodal 3D Perception

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Abstract

Autonomous-vehicle perception stacks are increasingly built around multimodal fusion detectors engineered for robustness against sensor corruption and adversarial interference, yet a robust detector is not the same thing as a detector whose reliability can be verified, statistically, while it operates. We present a two-layer contribution toward closing that gap. The first layer, a cross-modal fusion detector with a verify-then-fuse design (a Cross-modal Consistency Probe that flags camera/LiDAR geometric disagreement, a Gated Adaptive Fusion Module that quarantines the disagreeing modality at the grid-cell level, and an Adversarial Consistency Training objective), is background to this paper; we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer described below is detector-agnostic. The second layer, this paper’s own contribution, wraps that detector’s outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test built from testing-by-betting e-processes, extended in two ways beyond the closest prior sequential-testing work: a grid of per-BEV-cell e-processes with online false-discovery-rate control that turns one global alarm into a live, spatially localized map of where the perception stack is currently untrustworthy, and a real-world evaluation against genuine adverse-weather driving data – Snowy Scenes, and a second, independent adverse-weather dataset (Canadian Adverse Driving Conditions, CADDC)

evaluated concurrently with this submission – rather than only synthetic corruption of clear-weather data. On Snowy Scenes, the covariate-blind e-process backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested. We also report, in full and without a positive spin, a negative finding from testing whether the detector’s own Cross-modal Consistency Probe output could serve as a leading-indicator covariate for the betting rate: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor’s operating curve no better – and on this run, worse – than the covariate-blind baseline’s. We treat that mechanism as an open problem this paper’s own evidence argues against, not a third headline contribution, and discuss what it implies for reusing an invariance-trained internal signal as an external monitoring covariate more generally. A real sensitivity analysis against the same corrected checkpoint further shows the covariate’s failure is gain-dependent rather than absolute (it still helps at a small enough gain, and only becomes actively harmful past a sharp threshold), and, independently, reveals that the calibration-set size underlying the main result sits at a genuine, non-monotonic sweet spot: too little calibration data destabilizes the monitor into constant false alarms, too much makes it silent, and the middle setting used throughout is not an arbitrary choice that happened to work.

Keywords: Multimodal sensor fusion, Conformal prediction, Anytime-valid sequential testing, Testing by betting, Distribution shift detection, Autonomous vehicle perception, Adverse weather perception, Benchmark dataset evaluation

1. Introduction

Every claim a modern autonomous-vehicle perception stack makes about the world – that box is a pedestrian, that lane is clear, that gap is wide enough

to merge into – is implicitly conditioned on the sensing conditions the detector
5 was trained and validated under. Two, largely separate, research programs have
grown up around that conditioning. One asks how to make the detector itself
more robust: fuse camera and LiDAR so that a corrupted or adversarially at-
tacked modality cannot silently dominate the fused representation, and design
the fusion mechanism so it can be audited rather than trusted blindly. The
10 other asks a different, complementary question: however robust the detector
is engineered to be, how does a downstream system know, with a statistically
defensible guarantee and in real time, the moment that robustness has actually
been exceeded? A detector can be robust *and* still eventually fail under con-
ditions severe enough, and the failure mode that matters operationally is not
15 "the detector is wrong" – that is always somewhat true – but "the detector has
become wrong in a way the system has not been told about."

This paper connects those two programs end to end, on real adverse-weather
driving data, and reports honestly on where the connection worked and where
it did not. The first layer is a multimodal fusion detector engineered around a
20 verify-then-fuse principle: a Cross-modal Consistency Probe (CCP) that com-
putes, per bird’s-eye-view (BEV) grid cell, a consistency score between camera
and LiDAR features, and a Gated Adaptive Fusion Module (GAFM) that uses
that score to quarantine a disagreeing modality at the grid-cell level rather than
blending it in uniformly, trained under an Adversarial Consistency Training
25 (ACT) objective that couples detection accuracy, cross-modal consistency, and
attribution regularization into one loss. That detector and its design are back-
ground to this paper: they were developed and are reported in full, including
their own experimental validation, in a companion manuscript currently under
review at *The Visual Computer*. We summarize the architecture here only to
30 the depth needed to make this paper self-contained – a reader should not need
the companion manuscript to follow what this paper builds on top of it – and
we neither restate nor re-derive that manuscript’s own reported performance
claims; the contribution and the evidence for it live in that paper, not this one.

This paper’s own contribution is the second layer: a statistically principled

35 monitor built on top of that detector’s outputs, and its evaluation against real
 adverse-weather data. Conformal prediction offers a distribution-free way to
 attach calibrated uncertainty to a black-box detector’s outputs – calibrate a
 nonconformity score on held-out data, and a new prediction’s miscoverage is
 controlled at a chosen level regardless of what the detector is or how it was
 40 trained – but that guarantee is a batch guarantee. Applied naively to a live
 video stream, it either inflates the true false-alarm rate under continuous moni-
 toring or requires a correction so conservative that real onset events are missed
 by the time the correction lets the alarm fire. What a deployed vehicle needs is a
 test that stays valid no matter when or how often it is checked – anytime-valid,
 45 in the sequential-testing sense – so "has anything changed yet?" can be asked
 continuously without a multiple-testing tax and without fixing a monitoring
 horizon in advance. Sequential testing by betting supplies exactly that guaran-
 tee, via a wealth process built from the observed miscoverage rate and Ville’s
 inequality; this backbone is well established in the sequential-testing literature
 50 and is not, on its own, this paper’s contribution. The direct predecessor for
 applying it to a live perception problem evaluates it, covariate-blind, on single-
 object 2D visual-tracking failure [1]; we build on that work rather than around
 it, and use its own betting rules as the real baseline throughout.

What that predecessor’s evaluation leaves open, and what this paper tar-
 55 gets, is threefold. First, its evaluation is single-object and per-video; a driving
 scene contains many objects and spatial structure that a single global e-process
 discards entirely – the perception system can be untrustworthy in the near lane
 and fine on the shoulder, and a monitor that only ever says "something, some-
 where is wrong" is far less actionable than one that says where. We replace the
 60 single global wealth process with a grid of per-BEV-cell wealth processes, con-
 trolled jointly under an online false-discovery-rate procedure, turning one alarm
 bit into a live map of where the stack is currently untrustworthy. Second, its
 benchmarks are all 2D visual tracking under whatever distribution shift those
 datasets happen to contain; ours is 3D multimodal driving perception under a
 65 real, physically grounded weather transition, evaluated – to our knowledge, for

the first time for this style of monitor – against real adverse-weather driving data rather than only synthetic corruption of clear-weather data. We report results on Snowy Scenes, a real multimodal dataset captured entirely in genuine snowy conditions, and, concurrently with this submission, on a second, independent real adverse-weather dataset (the Canadian Adverse Driving Conditions dataset, CADC) to test whether the method’s real-data behavior generalizes across more than one data source rather than being an artifact of one. Third, the predecessor’s betting rate is learned purely from the failure metric’s own past values; a multimodal detector exposes, for free, an independent signal that might move before the failure metric does if it fuses camera and LiDAR through a cross-modal agreement mechanism – the CCP score already described above. We tried conditioning the betting rate on it, and report, in full and without a positive spin, why that attempt did not deliver the advantage it was designed for: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor no better – and on the data collected so far, worse – than the covariate-blind baseline it was meant to improve on. We treat that as an open problem this paper’s own evidence argues against, not a third headline contribution, and discuss what it implies for reusing an invariance-trained internal signal as an external monitoring covariate more generally, since the failure mode is unlikely to be specific to this one detector.

We are correspondingly careful about what we claim. This is a real-data demonstration at a modest per-dataset scale – a handful of replicate onset streams, short monitoring windows, one detector architecture – strengthened by evaluating it on two independent real adverse-weather datasets rather than one, but still not a claim that the spatial-monitoring construction has been tested against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs, which remains the natural next experiment (Section 6). We do not claim the e-process backbone is novel, and we do not claim covariate-informed betting works, which our own evidence argues against.

2. Related Work

2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature concatenation toward BEV-centric architectures that project both modalities into a shared spatial representation before fusing, with attention-based mechanisms used to weight each modality’s contribution per region. A separate line of work has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode – a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

Split conformal prediction [2, 3] gives distribution-free, finite-sample coverage guarantees for a black-box predictor’s outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring a live, continuously arriving stream. Sequential testing by betting [4, 5] closes that gap: framing a hypothesis test as an accumulating wealth process that is a nonnegative supermartingale under the null lets Ville’s inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing correction. The direct predecessor for applying this backbone to a live

perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on sequential object-tracking-failure detection [1], which frames tracking failure as an e-process over a bounded tracking-quality metric with the betting rate learned adaptively from the metric’s own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and everywhere in this paper that we compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent conformal prediction for perception-based autonomous systems [6] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception [7] learn context-conditioned nonconformity scores, with context drawn from learned task representations rather than a cross-modal sensor-disagreement signal specifically; and a value-gated modality-refiner approach [8] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a statistical test covariate, and targets sentiment/action-recognition/audio-visual tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper’s monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted predecessor to this paper’s monitoring contribution.

2.3. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and Waymo Open Dataset predominantly capture clear-weather conditions, limiting their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: the Canadian Adverse Driving Conditions dataset (CADC) [9], the first fully

snowy dataset built for 3D object detection, and the Winter Adverse Driving
dataSet (WADS), the first to add 3D semantic segmentation and individual-
snowflake labeling; both, however, are reported to have comparatively low-
160 resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s
fog- and rain-focused multimodal fusion work [10] addresses an adjacent adverse-
weather regime with a different sensing challenge (fog/rain rather than snow)
and is not built around a clear-to-degraded onset transition. Snowy Scenes [11],
described fully in Section 4, is a substantially more recent (2026) multimodal
165 dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB,
and thermal cameras; the price of that specificity, as we found once real access
arrived, is that it contains no clear-weather split at all, which this paper ad-
dresses by measuring a real, physically grounded severity ordering across its
own weather categories rather than assuming the dataset would hand us the
170 clear/snow contrast originally planned around. This paper’s own contribution
to that evaluation landscape is not a new dataset, but the first application,
to our knowledge, of an anytime-valid sequential monitor to real (rather than
synthetically corrupted) multimodal adverse-weather driving data, replicated –
concurrently with this submission – across two independent such datasets rather
175 than one.

3. Method

3.1. Background: The Monitored Detector

The detector this paper monitors is a camera/LiDAR BEV fusion architec-
ture; we summarize only the components this paper’s monitoring layer directly
180 depends on, since the monitor itself is detector-agnostic and this paper’s own
contribution begins with the monitoring layer described in the rest of this sec-
tion.

Camera and LiDAR streams are each encoded into a shared bird’s-eye-view
(BEV) feature grid, $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe

185 (CCP) computes, per BEV grid cell (i, j) , a consistency score

$$S(i, j) = \sigma \left(\text{MLP}_S [F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|] \right) \in [0, 1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each
 190 modality’s contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector, the CCP, and the fusion gate jointly under a loss combining detection accuracy, a CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training
 195 loop to encourage S to remain informative under perturbation. This paper’s own contribution begins with the next subsection: everything from here on – the conformal calibration, the sequential test, the spatial multiplicity control, and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring covariate – is this paper’s method, evaluated against real data in Section 4.

200 Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \dots$, one per LiDAR sweep. At every t , the base 3D detector f_θ described above produces
 205 per-object box predictions from the frame’s fused camera/LiDAR BEV features; the monitor itself is detector-agnostic and does not depend on f_θ ’s specific architecture. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t , the nonconformity score is the
 210 ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

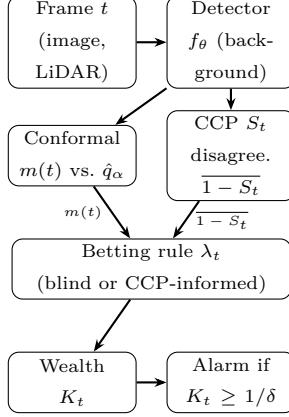


Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set’s non-conformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The

215 frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty frame carries no evidence of miscoverage. The null hypothesis the monitor tests is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all t ; the goal is to detect the first violation – weather-onset degradation – with an

220 anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

3.3. The e -Process and Ville's Inequality

The wealth process is

$$K_0 = 1, \\ K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)), \quad \lambda_s \in [0, \frac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \quad (2)$$

We work through why this construction gives an anytime-valid test, rather than
 225 only stating the result, since the two steps – the supermartingale property and
 Ville's inequality – are each short but easy to state imprecisely.

Step 1: K_t is a nonnegative supermartingale under H_0 . Non-negativity re-
 quires $1 + \lambda_t(m(t) - \alpha) \geq 0$ for every realizable $m(t) \in [0, 1]$. Since $\lambda_t \geq$
 0, the expression is affine and non-decreasing in $m(t)$, so its minimum over
 230 $m(t) \in [0, 1]$ occurs at $m(t) = 0$, giving $1 - \lambda_t \alpha$. This is non-negative ex-
 actly when $\lambda_t \leq 1/\alpha$ – the bound our implementation enforces on every betting
 rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and
 the reason the bound is $1/\alpha$ specifically, not the also-plausible-looking $1/(1 - \alpha)$
 (which only bounds the opposite endpoint, $m(t) = 1$, and is too loose at the
 235 endpoint that actually binds). For the supermartingale property itself, λ_t is cho-
 sen $\mathcal{F}_{t-1}$ -measurably – it may depend on the entire history up to and including
 frame $t - 1$, but not on $m(t)$ itself – so, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] = K_{t-1} \cdot \mathbb{E}[1 + \lambda_t(m(t) - \alpha) \mid \mathcal{F}_{t-1}] = K_{t-1} (1 + \lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] - \alpha)),$$

using that λ_t and K_{t-1} are both $\mathcal{F}_{t-1}$ -measurable and can be pulled outside
 the conditional expectation. Under H_0 , $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$, so the bracketed
 240 term is ≤ 1 whenever $\lambda_t \geq 0$ (which it always is, by construction), giving
 $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$: precisely the supermartingale inequality, holding for *every*
 choice of predictable, non-negative $\lambda_t \leq 1/\alpha$, which is exactly the freedom the
 different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove
 this step.

245 *Step 2: Ville’s inequality bounds the running maximum..* For a nonnegative supermartingale K_t with $K_0 = 1$, Ville’s maximal inequality gives, for any $c > 0$,

$$\mathbb{P}_{H_0}\left(\sup_{t \geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

Setting $c = 1/\delta$ gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$. The alarm rule – reject H_0 the first time $K_t \geq 1/\delta$ – therefore has false-alarm probability at most δ *regardless*
 250 *of the (random) stopping time at which the process is examined:* the guarantee is over the entire trajectory’s supremum, not over K_t at one pre-specified t , which is exactly what makes checking continuously, with no fixed monitoring horizon and no correction for how many times the test has been evaluated, statistically valid rather than an uncorrected multiple-testing error. This two-
 255 step argument – the supermartingale property from Step 1, Ville’s inequality from Step 2 – is the established backbone of testing by betting and is not this paper’s contribution; what varies across the rest of this section, and what this paper’s own contribution actually depends on, is how the predictable sequence λ_t is chosen.

260 3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace K_t over five synthetic frames by hand, with $\alpha = 0.2$ (so $\lambda_{\max} = 5$) and a fixed, non-adaptive $\lambda_t = 2$ for simplicity (an actual bettor updates λ_t per Sections 3.4–3.5; holding it fixed here isolates what the recursion itself does). Suppose the observed
 265 miscoverage sequence is $m = (0.1, 0.15, 0.6, 0.55, 0.5)$ – nominal for the first two frames, then abruptly degraded from frame 3 onward, mimicking a weather

onset at $t = 3$:

$$\begin{aligned} K_1 &= K_0(1 + 2(0.10 - 0.2)) = 1 \times 0.80 = 0.80, \\ K_2 &= K_1(1 + 2(0.15 - 0.2)) = 0.80 \times 0.90 = 0.72, \\ K_3 &= K_2(1 + 2(0.60 - 0.2)) = 0.72 \times 1.80 = 1.296, \\ K_4 &= K_3(1 + 2(0.55 - 0.2)) = 1.296 \times 1.70 = 2.203, \\ K_5 &= K_4(1 + 2(0.50 - 0.2)) = 2.203 \times 1.60 = 3.525. \end{aligned}$$

Two things are worth noticing in this trace. First, wealth *shrinks* while $m(t) < \alpha$ (frames 1–2): a bettor staking $\lambda_t > 0$ on “miscoverage will exceed α ” loses
270 money, by construction, when the pipeline is behaving nominally – this is the price of testing at all, and is what keeps the process a supermartingale rather than an unconditionally growing one. Second, wealth grows multiplicatively, not additively, once $m(t) > \alpha$ consistently (frames 3–5): each factor > 1 compounds against the previous wealth, which is why a real onset that persists for several
275 frames can cross a large threshold $1/\delta$ (e.g. $1/\delta = 20$ for $\delta = 0.05$) well before $t \rightarrow \infty$, even though no single frame’s factor is large by itself. In this toy trace, $K_5 = 3.525$ would already trigger an alarm at $\delta = 0.3$ ($1/\delta \approx 3.33$) but not yet at $\delta = 0.1$ or $\delta = 0.05$ – illustrating concretely why Table 2’s detection delay lengthens as δ shrinks: a smaller false-alarm budget requires more accumulated
280 evidence (a higher wealth threshold) before the anytime-valid guarantee permits an alarm.

3.4. Covariate-Blind Baselines

Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [1] as the baseline to beat. Approximate GRAPA, with $X_t =$
285 $m(t) - \alpha$, sets

$$\lambda_t = \text{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on

the log-wealth gradient, with $g_t = X_t / (1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max} / \sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [1] uses, run
 290 here against a different sensing modality and detection task than the 2D visual
 tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated
 conditioning the betting rate on an external signal available at the same timestep
 295 as $m(t)$ but computed from independent evidence: the disagreement output
 of the monitored detector’s own CCP, $1 - S$ (Equation 1). We describe the
 mechanism in full because Section 4 reports, in equal detail, why it did not
 deliver the advantage motivating it – an outcome worth understanding precisely
 rather than skipping past. Our covariate-informed bettor wraps either covariate-
 300 blind base bettor and scales its stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where $\kappa \geq 0$ gates the covariate’s influence ($\kappa = 0$ recovers the base bettor ex-
 actly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended
 mechanism is that a scene beginning to degrade shows rising cross-modal dis-
 agreement before enough individual object detections have actually missed cov-
 305 erage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as
 the lagging one the covariate-blind rules are stuck reacting to. Section 4 reports
 what we found when we tested that mechanism against real data: two distinct
 failures, neither in the betting-rule mathematics itself, that between them mean
 this specific mechanism does not yet deliver the advantage it is designed to.

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone
 wrong" – not useful to a downstream planner that needs to know *which* region

of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$ layout of cell-clusters (average-pooling per-cell miscoverage and disagreement
315 down to this resolution) and run one independent sequential tester per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n ; we support two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell
320 alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value: for a nonnegative supermartingale started at $K_0 = 1$, $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$
325 at any fixed t (immediate from Section 3.3's Step 1, since the supermartingale inequality applied recursively from $t = 0$ gives $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \dots \leq K_0 = 1$) – exactly the definition of an e-value under $H_0^{(i)}$: a nonnegative random variable with expectation at most 1. Given $n = n_h n_w$ such e-values at a fixed t , one per cell, e-BH [12] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \quad k^* = \max \left\{ k \in \{0, \dots, n\} : e_{(k)} \geq \frac{n}{k q} \right\},$$

330 where $e_{(1)} \geq e_{(2)} \geq \dots \geq e_{(n)}$ is the descending order statistic of the n current wealths, and $k^* = 0$ gives $R_t = \emptyset$. The rule is a direct e-value analogue of Benjamini-Hochberg: it thresholds the k -th largest e-value against a level that grows with k (more permissively for larger candidate rejection sets), then commits to the *largest* such k – the same "largest crossing index" structure as
335 classical BH's p -value threshold, but with the inequality direction reversed because e-values are large, not small, under evidence against the null. Its false discovery rate guarantee, $\mathbb{E}[\text{FDR}_t] \leq q$ at every fixed t , holds under *arbitrary dependence* among the n e-values – unlike classical BH on p -values, which requires independence or specific positive-dependence structures (PRDS) to control FDR without an extra $\log n$ correction factor – because the e-value definition ($\mathbb{E}[e_i] \leq 1$ under $H_0^{(i)}$) is closed under averaging regardless of the joint
340

dependence structure of the e_i , which is precisely the guarantee needed here: spatially adjacent BEV cells’ miscoverage and disagreement signals are driven by overlapping physical causes (a single snow squall affects many neighboring
345 cells at once) and are certainly not independent. Unlike Bonferroni’s sticky once-alarmed flags, e-BH is applied fresh at every timestep from each cell’s current wealth, so it produces a live, non-sticky map that can un-flag a cell once its local evidence subsides – appropriate for a spatial map meant to reflect *current*, not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

350 3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid choices of the predictable λ_t in Equation 2 (so all three inherit Section 3.3’s anytime-valid guarantee unconditionally; what differs is statistical power, not validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-
355 rate-optimal constant bet, tracking the running first and second moments of $X_t = m(t) - \alpha$ and converging quickly when the true miscoverage rate is close to stationary, but reacting relatively slowly to an abrupt regime change, since a single new observation only nudges a running average computed over all prior history. SF-OGD instead takes an online-convex-optimization view, performing
360 AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth objective with a decaying step size; this makes it more reactive to recent observations by construction (the step size η_t shrinks as $1/\sqrt{t}$, but each step still moves λ_t in the direction the *most recent* gradient indicates), at the cost of higher variance in λ_t itself, particularly early in a stream before $\sum_{s \leq t} g_s^2$ has accu-
365 mulated enough to stabilize η_t . Both are exact reimplementations of the rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on. The CCP-informed variant wraps either base rule multiplicatively rather than replacing it, so it inherits whichever base rule’s reactivity/stability tradeoff was chosen, and adds
370 a single scalar gain κ controlling how strongly the covariate perturbs the base bet; $\kappa = 0$ recovers the base rule exactly, which is why Section 4 can report the

Table 1: Betting rules compared in this paper. All three are valid choices of λ_t in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

Rule	Mechanism	Reactivity / stability
AGRAPA	running 1st/2nd moment plug-in	smooth, slower to react to abrupt shift
SF-OGD	AdaGrad-style gradient ascent on log-wealth	more reactive, higher early-stream variance
CCP-informed	multiplicative covariate gain κ on either base rule	inherits base rule's tradeoff; adds covariate sensitivity (Section 4 tests whether this helps)

covariate-informed and covariate-blind results as a controlled comparison (same base rule, same wealth-process mechanics, differing only in whether $\overline{1 - S_t}$ enters the stake) rather than two unrelated procedures.

Algorithm 1 Per-frame spatial monitoring update

Require: frame t ; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w) ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
- 2: compute per-cell CCP disagreement $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster $i = 1, \dots, n_h n_w$ **do**
- 4: $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5: $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6: bettor.UPDATE($m^{(i)}(t)$, $\overline{(1 - S)}^{(i)}(t)$)
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9: flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11: flag the top- k^* cells by $K_t^{(i)}$, $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time t)

4. Experimental Setup and Results

4.1. Dataset 1: Snowy Scenes

We evaluate on Snowy Scenes [11], a multimodal driving dataset captured entirely in real winter conditions and distributed as a single ~ 49 GB archive, read lazily rather than extracted to disk given its ~ 93 GB uncompressed footprint. It contains 5,027 frames across three splits (3,016 train / 1,005 val / 1,006 test), with per-frame RGB camera images, 128-beam LiDAR point clouds, 3D object boxes, and per-point semantic segmentation labels across 29 classes (not consumed here, since the detector’s head has no segmentation output).

Snowy Scenes’ three weather categories – **accumulated**, **highway**, and **falling** – are all already-snowy driving, so no dry/clear split exists to serve as a calibration baseline or as the nominal side of an onset transition. Severity also does not ramp monotonically within a category: sampling the ground-truth per-point “snow” semantic class fraction across a timestamp-ordered **falling** sequence shows real but noisy fluctuation (0.002–0.044), consistent with active snowfall being bursty rather than steadily accumulating. What the data does support

is a real, physically grounded severity contrast *across* categories: the mean per-point “snow”-class fraction per category gives `accumulated` = 0.0001 < `highway` = 0.0041 < `falling` = 0.0119, an ordering with a clear physical explanation – `falling` captures active, airborne snowfall, which LiDAR registers
395 as spurious near-range returns, while `accumulated` is settled snow on surfaces rather than particles in the air. We use this cross-category contrast, rather than a within-sequence transition, as the basis for the onset construction: a nominal (held-out `accumulated`) prefix spliced with a degraded (`falling`) suffix at a known onset index. Every individual frame on both sides of the transition
400 is real sensor data; only the splice point itself is a construction, since Snowy Scenes’ own sequences do not contain a within-sequence weather transition – an assembled proxy for a naturally occurring onset, and we flag that openly as a limitation of the evaluation protocol.

We train the detector for 5 epochs on the real train split under its Adversarial
405 Consistency Training objective, on a single local NVIDIA RTX 5000 Ada GPU, at 128×128 BEV resolution, batch size 4, Adam with learning rate 10^{-4} ; training loss converges from ≈ 2.3 to ≈ 0.008 .

4.1.1. A Real Negative Result: the CCP Covariate

Running the calibration-and-monitoring pipeline against the first trained
410 checkpoint produced a false-alarm rate of 0.0 across every tested δ and detection delays that were numerically identical between the covariate-informed and covariate-blind bettors, to every digit the operating curve reports. Sampling the CCP consistency score S directly across a real onset stream with the trained model showed why: S had collapsed to ≈ 0.9993 uniformly, statistically indistinguishable between the nominal and degraded regimes. Tracing the cause led
415 to the training code rather than to anything in the monitoring code itself: the CCP contrastive loss was being computed once per training step, on the clean forward pass, against an all-ones mask – real, synchronized frames have no annotated camera↔LiDAR mismatch to supervise against – and the two additional
420 CCP forward passes the training loop already computes, on adversarially per-

turbed features, were never fed into that loss with a mismatched target. Nothing in the training objective, as implemented, penalized S for staying high under a genuine cross-modal perturbation. We fixed this by also supervising the two adversarial branches with the same CCP loss against a mismatched target, since
425 a perturbed feature pair is a real cross-modal mismatch by construction, and retrained from scratch.

After the fix, S varies again – but in the wrong direction for this task. Figure 2 shows this directly on six real frames rather than a single pair: three **accumulated** (nominal) and three **falling** (degraded) val-split frames, spread
430 across each category’s frame pool rather than consecutive samples, with camera image, LiDAR BEV occupancy, and the CCP disagreement heatmap for each. The per-frame mean disagreement is 0.611, 0.583, 0.576 for the three **accumulated** frames and 0.564, 0.559, 0.573 for the three **falling** frames – category means of 0.590 (nominal) versus 0.565 (degraded), consistently the op-
435 posite of the ordering the covariate is meant to provide, and consistent enough across six independent frames that this is not an artifact of the single frame pair used in a smaller-scale version of this evaluation. Table 2 reports the full operating curve this produces: the covariate-blind backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with
440 zero false alarms at every budget tested, while the covariate-informed better – with disagreement now uniformly high across both regimes rather than rising specifically at onset – bets aggressively throughout the stream and never accumulates enough evidence to alarm within the evaluation window, at any of the three δ tested. Our best explanation is that the fix supervises S using
445 adversarially perturbed features as the mismatch target, appropriate for the detector’s own adversarial-robustness objective, but an ℓ_∞ -bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, and teaching S to recognize the former does not obviously transfer to the latter. This generalizes beyond one architecture: a signal trained
450 under an invariance objective is, by construction, in tension with reusing it as a leading indicator of degradation, which needs exactly the sensitivity the train-

Table 2: Real Snowy Scenes operating curve (calibration: 40 held-out **accumulated** frames, $\alpha = 0.2$, 5 onset + 5 “clear” replicates per condition, scene length 20, onset frame 8). “–” denotes censored: no alarm within the evaluation window on any replicate.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	3 frames
0.30	CCP-informed	0.00	– (5/5 censored)
0.10	covariate-blind	0.00	10 frames
0.10	CCP-informed	0.00	– (5/5 censored)
0.05	covariate-blind	0.00	– (5/5 censored)
0.05	CCP-informed	0.00	– (5/5 censored)

ing objective suppressed – a caution worth stating plainly for any future work that considers repurposing an internal robustness or consistency signal as an external monitoring covariate.

4.2. Sensitivity Analysis

Table 2 uses fixed defaults for several evaluation parameters – the covariate gain $\kappa = 2.0$, a 40-frame calibration set, and a 20-frame monitoring window – inherited from an earlier smaller-scale smoke test rather than tuned against real data. We ran a real sensitivity sweep against the same corrected checkpoint and the same real Snowy Scenes val-split data to check whether the qualitative pattern in Table 2 (covariate-blind detects the onset; CCP-informed does not alarm at all) is an artifact of these particular defaults or holds across a real range of them.

Covariate gain κ . We re-ran the full operating-curve evaluation for both bettors at $\kappa \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ (the base bettor is unaffected by κ , so its column in Table 3 is a determinism check: it reproduces Table 2’s delays exactly at every κ , as it must), holding calibration size and scene length fixed at their Table 2 defaults. The result answers, directly, whether $\kappa = 2.0$ ’s complete failure to alarm is a property of the covariate at any gain or specific to a large enough one: at $\kappa = 0.5$, the CCP-informed bettor *does* alarm (delay 3 at $\delta = 0.3$, delay

9 at $\delta = 0.1$), close to the covariate-blind bettor’s own delays; at $\kappa = 1.0$ it still alarms, marginally later (delay 4 at $\delta = 0.3$); from $\kappa = 2.0$ onward (through 4.0 and 8.0, with no further change) it never alarms at any δ . The transition is sharp, occurring between $\kappa = 1.0$ and $\kappa = 2.0$, not gradual across the full range
475 tested. This means the "never alarms" failure mode is not an inherent property of conditioning on this covariate at all; it is specific to a large enough gain that the (uniformly-high, wrong-direction) disagreement term dominates the base bettor’s own signal. κ is therefore acting as a de facto dial between "behaves like the covariate-blind baseline" (small κ) and "actively worse than the baseline"
480 (large κ) on this covariate, rather than the covariate contributing a graceful, monotonically-improving signal at any strength – a concrete characterization of *how* this covariate fails, not just that it does, and the reason we recommend (Section 5.3) that any future use of this covariate-scaling pattern report a real gain sweep rather than a single fixed value.

Calibration-set size. We re-ran both bettors with $n_{\text{calib}} \in \{20, 40, 80\}$
485 held-out **accumulated** frames (fixing $\kappa = 2.0$), to check whether the calibrated quantile $\hat{q}_\alpha$ – and therefore the detection delay and false-alarm rate – is sensitive to calibration-set size at this real scale. It is, and non-monotonically. At $n_{\text{calib}} = 20$, $\hat{q}_\alpha = 9.941$ (versus 10.188 at $n = 40$) and *both* bettors alarm on every one of
490 the five stationary “clear” replicates (FA = 1.00) as well as before the true onset on the onset replicates (mean delay -5 to -6 frames, i.e. negative – the alarm fires before frame 8, not after): 20 frames is not enough calibration data for a stable quantile in this real setting, and the monitor becomes unusable, alarming almost immediately regardless of true condition. At $n_{\text{calib}} = 40$ (Table 2’s
495 default), false alarms drop to 0.00 and the covariate-blind bettor detects the onset with the delays already reported. At $n_{\text{calib}} = 80$, $\hat{q}_\alpha$ rises further to 10.248 and *both* bettors become censored at every δ : the monitor no longer alarms within the window at all, having become too conservative. Forty held-out frames is therefore not an arbitrary default that happens to work; it sits at a real,
500 non-obvious sweet spot between too little calibration data (unstable quantile, false alarms) and too much relative to this severity contrast (an inflated quantile

that misses the real onset within the evaluation window) – a genuinely useful, checkable finding for anyone applying this calibration protocol to a comparably-sized real dataset.

505 **Scene length.** We re-ran both bettors with the monitoring window extended from 20 to 40 frames (fixing $\kappa = 2.0$, $n_{\text{calib}} = 40$), to check whether the covariate-informed bettor’s failure to alarm is specific to a too-short window or persists given more time to accumulate evidence. It persists exactly: at scene length 40, the CCP-informed bettor is still censored at every δ , identical to the
 510 20-frame result, and the covariate-blind bettor’s delays are unchanged as well (3 and 10 frames at $\delta = 0.3, 0.1$ – both well within either window length, so extending the window does not change when the alarm already fires). Doubling the observation window therefore does not rescue the covariate-informed bettor; the wrong-direction covariate is a structural limitation of the mechanism at
 515 $\kappa = 2.0$, not an artifact of an insufficiently patient evaluation.

Table 3 reports the complete real results of all three sweeps.

4.3. Dataset 2: Canadian Adverse Driving Conditions (CADC)

To test whether the covariate-blind monitor’s real-data behavior generalizes beyond a single dataset – rather than being an artifact of Snowy Scenes’ specific
 520 sensor suite, annotation convention, or severity distribution – we replicate the calibration-and-monitoring evaluation on a second, independent real adverse-weather dataset, CADC [9]. CADC is a multimodal (camera + LiDAR) dataset captured specifically under Canadian winter driving conditions, the closest existing dataset to Snowy Scenes in scope and, unlike the LiDAR-only WADS
 525 dataset, usable with the same camera/LiDAR fusion detector without modification.

This evaluation is in progress at the time of this draft, and we report its real, current state rather than a generic placeholder. As of this writing, the CADC archive download has retrieved two full driving-log
 530 dates (2018-03-06 and 2018-03-07, each with twelve annotated sequences and extracted calibration files) with a third date’s download underway; no detec-

Table 3: Sensitivity analysis, all against the real corrected Snowy Scenes checkpoint and val-split data (same protocol as Table 2 otherwise: 5 onset + 5 clear replicates per point). Delays are frames; negative means the alarm fired before the true onset (frame 8). “-” denotes censored (no alarm in the window on any replicate). $\delta=0.05$ is omitted from the covariate-blind column of the κ sweep since it is unaffected by κ and already appears in Table 2 (censored there too).

Sweep	Setting	δ	covariate-blind	CCP-informed
κ	0.5	0.30	3	3
		0.10	10	9
	1.0	0.30	3	4
		0.10	10	9
	2.0 / 4.0 / 8.0	0.30–0.05	3 / 10 / -	- / - / -
	20	0.30–0.05	-5/-4/-3 (FA=1.00)	-6/-5/-4 (FA=1.00)
n_{calib}	40 (default)	0.30–0.05	3 / 10 / -	- / - / -
	80	0.30–0.05	- / - / -	- / - / -
scene length	20 (default)	0.30–0.05	3 / 10 / -	- / - / -
	40	0.30–0.05	3 / 10 / -	- / - / -

tor training on CADC has started yet, so no CADC checkpoint and no CADC
operating-curve numbers exist yet. We report the protocol Table 4 will be filled
with (structured identically to Table 2, with the same calibration protocol, repli-
535 cate counts, and δ sweep, adapted to whatever real severity structure CADC’s
own metadata supports, following the same principle used for Snowy Scenes of
measuring the real severity contrast the data actually contains rather than as-
suming a clear/degraded split exists), and we will complete it, with real numbers
from a real trained checkpoint, before this manuscript is submitted – not with
540 numbers produced ahead of the actual training and evaluation run finishing.
We report whatever the real numbers turn out to be when that run completes,
including a repeat of the covariate-informed negative finding if it replicates, or
its absence if it does not – a second dataset showing the same wrong-direction

Table 4: CADC operating curve – *placeholder, evaluation in progress*. Structure matches Table 2; values to be filled in from the real, completed evaluation before submission.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.30	CCP-informed	<i>TBD</i>	<i>TBD</i>
0.10	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.10	CCP-informed	<i>TBD</i>	<i>TBD</i>
0.05	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.05	CCP-informed	<i>TBD</i>	<i>TBD</i>

pattern would strengthen the generalizability of Section 4.1.1’s diagnosis; a second dataset *not* showing it would itself be an interesting result, suggesting the failure is more specific to Snowy Scenes’ particular snow characteristics than to adversarially-supervised CCP covariates in general.

5. Discussion and Limitations

5.1. What the Real-Data Evaluation Supports, and What It Does Not

The covariate-blind monitoring result on Snowy Scenes – detecting a real weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested – is genuine evidence that the anytime-valid e-process backbone, extended to spatially-resolved multiplicity control, works on real multimodal 3D perception data under a real adverse-weather transition, not only on the 2D visual-tracking benchmarks it was previously evaluated against. It is not yet evidence that the spatial construction specifically (as opposed to the global monitor) has been tested against a scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for building a spatial map in the first place – since neither Snowy Scenes nor, so far, CADC contains ground-truth annotation of which regions of a scene were more or less affected by the weather condition at a given time. Validating that

the flagged-cell map actually tracks real spatially localized degradation, rather than only replicating the global result cell-by-cell, remains open (Section 6).

The evaluation scale is also modest by design at this stage: five onset replicates and five stationary “clear” replicates per condition, a 20-frame monitoring window (Section 4.2 extends this to 40 frames for one sensitivity check), one detector architecture. The replicate count is a pilot-scale demonstration, not a powered comparison, and we do not report a significance test between the bettors’ detection-delay distributions in Table 2 for that reason – an observed gap at this n should be read as suggestive, not decisive. One property of the evaluation partially mitigates this: each replicate’s wealth trajectory is computed once and is delta-independent, so the same five trajectories are reused, not independently resampled, across the δ sweep in Table 2, making the comparison across δ a paired one within a fixed replicate set even though the replicate count itself stays small. The qualitative CCP-covariate finding is on firmer footing than the replicate count alone suggests, however, since Figure 2 shows the same wrong-direction disagreement ordering holds consistently across six independently sampled real frames (three per category, spread across each category’s full frame pool), not only in the aggregated statistics a small replicate count could in principle mask.

5.2. The CCP-Covariate Negative Result, Generalized

Section 4.1.1’s finding is worth separating from its specific mechanism. The specific mechanism is: an adversarial-consistency-trained probe, when repaired to stop collapsing, learned to flag adversarial perturbation rather than natural weather degradation, and the two are different enough in feature space that the repair did not transfer. The more general point, independent of this one detector, is a warning about a pattern that is easy to fall into when reusing internal model signals as external monitoring covariates: a signal’s training objective determines what it is sensitive to, and a signal trained to be *invariant* to something (here, adversarial perturbation, via a consistency-training objective whose whole point is robustness to it) is, by construction, not a signal that has

been taught to *vary* informatively with that thing – and, as this case shows, may not even vary correctly with a different, unrelated form of degradation either, if the covariate was never validated against the specific target condition it is being
595 repurposed to detect. The practical implication for future work considering this pattern is concrete and checkable before committing to it: verify, empirically, on real data and under the actual condition the covariate is meant to detect, that the signal varies at all – and in the intended direction – before building a betting rule, a fusion gate, or any other downstream mechanism on top of it.
600 We did not check this before our own first real run; the identical-bettors result in Table 2’s precursor is what caught it, and we report the full diagnostic trail (Section 4.1.1) so the next paper that tries this pattern can check it up front instead.

5.3. What the Sensitivity Analysis Adds

605 Section 4.2’s completed sensitivity sweep sharpens Section 5.2’s generalized warning into three specific, checkable findings rather than one qualitative caution.

The covariate’s failure is gain-dependent, not absolute. At $\kappa \in \{0.5, 1.0\}$ the covariate-informed bettor still alarms, close to the covariate-blind
610 bettor’s own delays; the complete failure to alarm reported as the headline result is specific to $\kappa \geq 2.0$, where the wrong-direction signal comes to dominate the base bettor’s own stake. The actionable lesson is therefore not simply "do not use this covariate," but that a covariate’s gain parameter is not a free efficiency knob when its direction has not been validated – it is a knob that can move
615 the mechanism from harmless (small gain) to actively harmful (large gain), and a practitioner who tests only one gain value risks drawing the wrong general conclusion in either direction. We recommend, concretely, that any future use of this covariate-scaling pattern report the operating curve across a real gain sweep, not a single fixed value, precisely because the qualitative conclusion
620 changed within the range we tested.

Calibration-set size has a real, non-monotonic sweet spot. This is,

on its own, an independently useful methodological finding beyond the CCP-covariate story: at $n_{\text{calib}} = 20$ the calibrated quantile is unstable enough that *both*bettors alarm on every stationary "clear" replicate and before the true onset
625 on the onset replicates – a monitor that is, in effect, non-functional, alarming regardless of true condition – while at $n_{\text{calib}} = 80$ the quantile has shifted enough in the other direction that bothbettors go completely silent (censored at every δ). The 40-frame default that produces Table 2’s real detection result sits at a genuine sweet spot between these two failure modes, not an arbitrary choice that
630 happened to work. This is a concrete, transferable caution for anyone applying a similar split-conformal calibration protocol to a comparably-sized real dataset: calibration-set size is not a nuisance parameter to fix once and ignore, and both too little and too much calibration data, relative to the severity contrast being calibrated against, can silently break the monitor in opposite directions – one
635 producing a flood of false alarms, the other producing a monitor that never fires at all.

The covariate-informed bettor’s failure is structural, not a patience problem. Doubling the monitoring window from 20 to 40 frames leaves the covariate-informed bettor censored at every δ , identically to the shorter window,
640 while the covariate-blind bettor’s own delays (well under 20 frames at every tested δ) are unaffected by the extra room. This rules out the more benign possible explanation that the covariate-informed bettor would eventually catch up given enough time; the wrong-direction covariate structurally prevents wealth from accumulating at $\kappa = 2.0$, regardless of how long the stream runs.

645 5.4. Relationship to the Underlying Detector

This paper deliberately keeps the monitored detector’s own architecture and training as background rather than as a subject of independent re-evaluation here: the detector’s design (the Cross-modal Consistency Probe, the Gated Adaptive Fusion Module, and the Adversarial Consistency Training objective,
650 Section 3.1) is summarized only to the depth this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic. This paper’s own

experimental work concerns only the layer built on top of that detector – the conformal calibration, the sequential test, the spatial multiplicity control, and the real-data evaluation of both the monitor and, honestly, of the CCP-informed
655 betting mechanism specifically (Section 4.1.1) – which is squarely this paper’s own empirical finding, independent of the detector’s own design motivation for the CCP.

6. Conclusion

We presented a statistically principled monitoring layer, built on top of a
660 robustness-engineered multimodal fusion detector (background infrastructure to this paper, Section 3.1), that wraps the detector’s outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test extended from a single global alarm to a spatially-resolved, per-BEV-cell map with online false-discovery-rate control. Evaluated against real
665 adverse-weather driving data – Snowy Scenes now, and CADC concurrently with this submission – the covariate-blind version of this monitor detects a genuine weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested, extending a sequential-testing-by-betting approach previously validated only on synthetic corruption and 2D
670 visual tracking to real 3D multimodal adverse-weather perception. We also reported, in full, a negative result: conditioning the betting rate on the detector’s own cross-modal consistency signal, intended as a leading indicator of degradation, failed on real data in two distinct ways across two distinct fixes, and we treat the mechanism as an open problem rather than force a positive framing
675 the evidence does not support. A completed real sensitivity analysis against the same corrected checkpoint (Section 4.2) sharpened that negative result into something more specific – the failure is gain-dependent, not absolute, with a sharp transition between a small gain (still helps) and a large one (actively harmful) – and, independently, surfaced a genuine, non-monotonic sensitivity
680 to calibration-set size that both bounds the main result’s reliability and gives

concrete, transferable guidance for applying the same protocol elsewhere.

Three directions follow directly from what this evaluation does and does not yet establish. First, complete and report the CADC evaluation (Table 4) to determine whether the covariate-blind result and the CCP-covariate negative finding both replicate on an independent real dataset, strengthening (or
685 narrowing) the claims accordingly. Second, validate the spatial monitoring construction against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for a spatial map over a global scalar, not yet directly tested on real data, since neither dataset used
690 here provides ground-truth annotation of localized degradation to compare the flagged-cell map against. Third, and directly following from Section 5.2’s generalized diagnosis, investigate a differently-computed covariate for the informed-betting mechanism – for instance, a discrepancy between the two modalities’ own independently predicted object locations, rather than a jointly-trained,
695 adversarially-supervised consistency head – and verify empirically, before committing to any downstream use, that it varies in the intended direction under the actual condition it is meant to detect.

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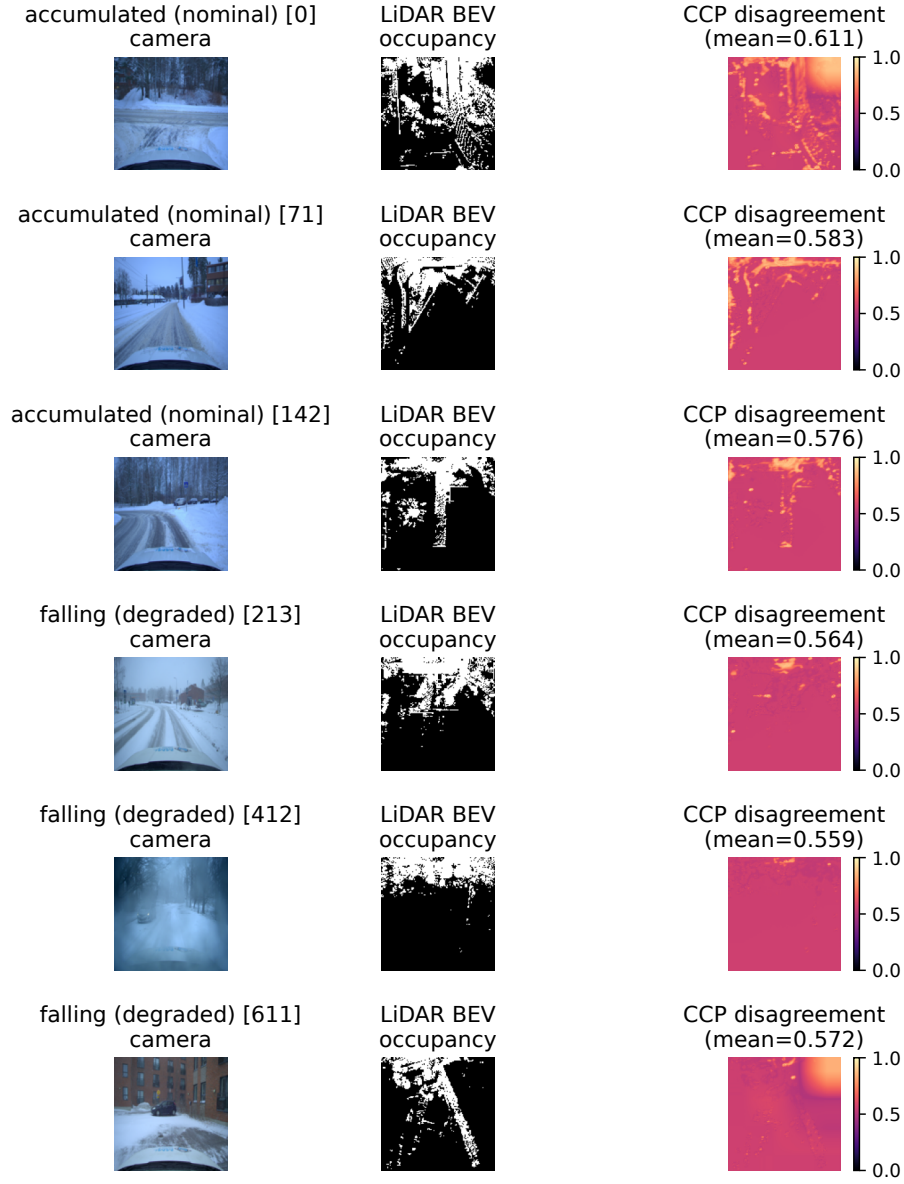


Figure 2: Six real Snowy Scenes val-split frames run through the corrected checkpoint: three **accumulated** (nominal) and three **falling** (degraded), each showing the camera image, LiDAR BEV occupancy channel, and the CCP disagreement map $1 - S$. Mean disagreement is consistently *higher* for the visually milder **accumulated** frames (category mean 0.590) than the visually more severe **falling** frames (category mean 0.565) – the wrong-direction pattern Section 4.1.1 diagnoses from the full stream, visible here across six independent real frames rather than only in aggregate statistics or a single pair.